In [10]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [2]:
```
df=sns.load_dataset("titanic")
df.head()
```

Out[2]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T

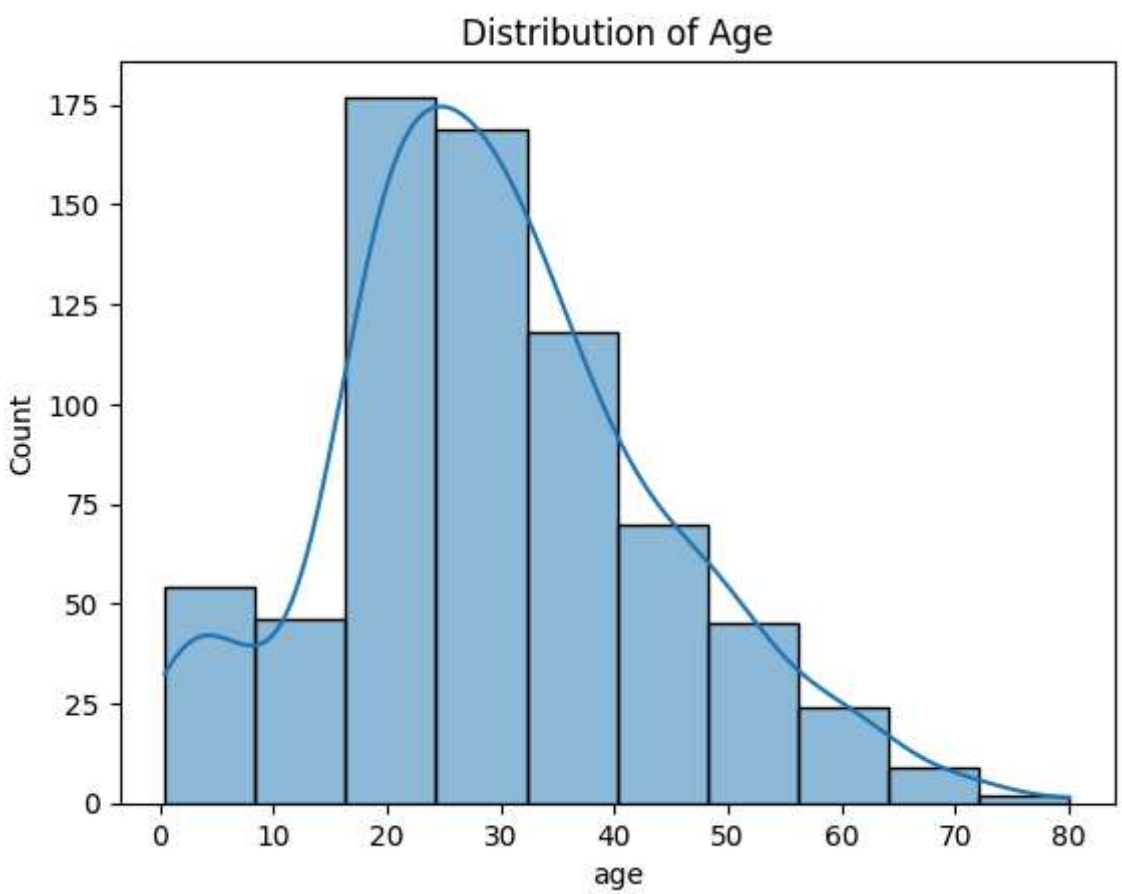
In [3]:
```
df.info()
df.describe()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    str
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    str
 8   class        891 non-null    category
 9   who          891 non-null    str
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    str
 13  alive        891 non-null    str
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 80.7 KB
```

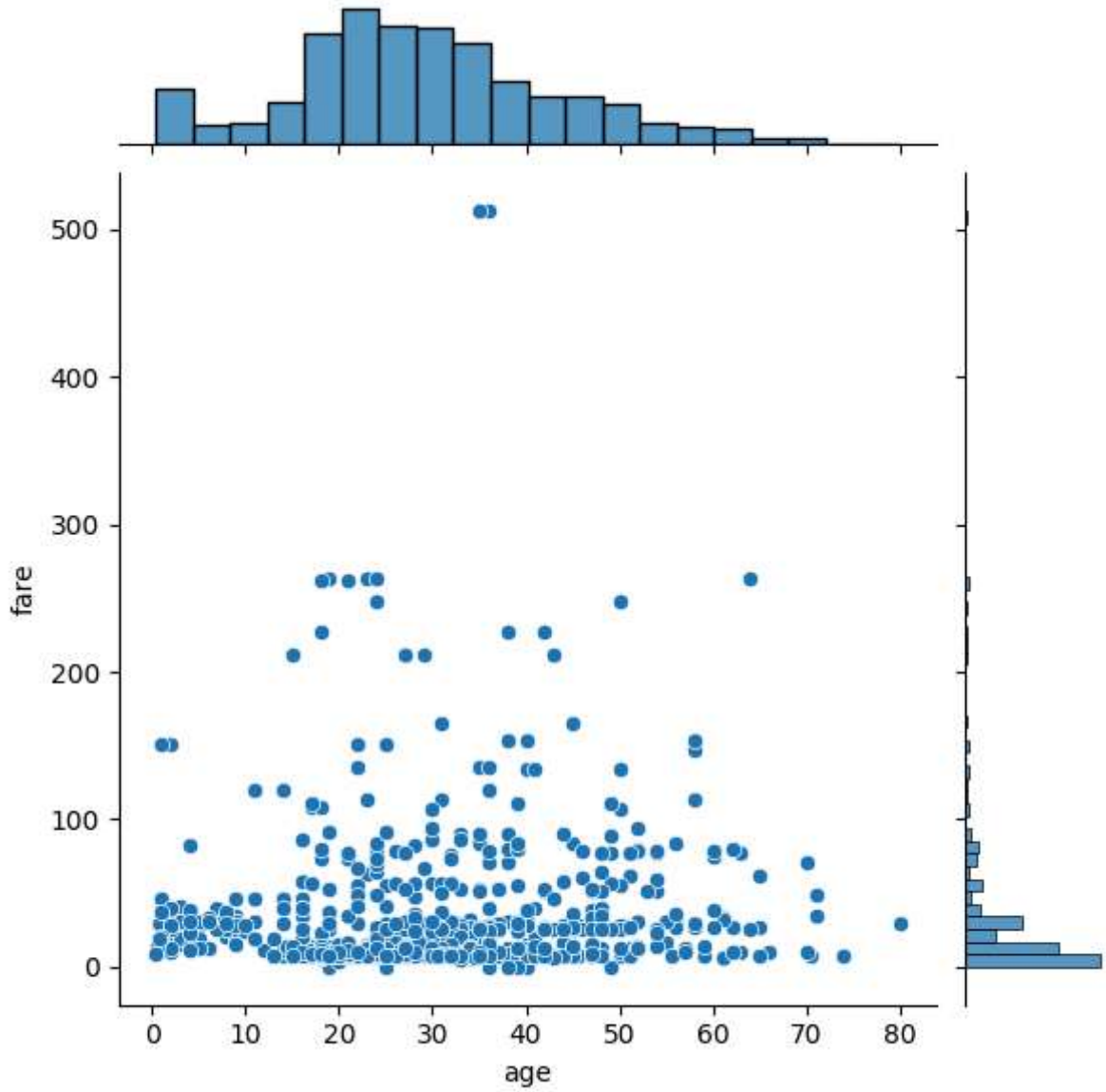
Out[3]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

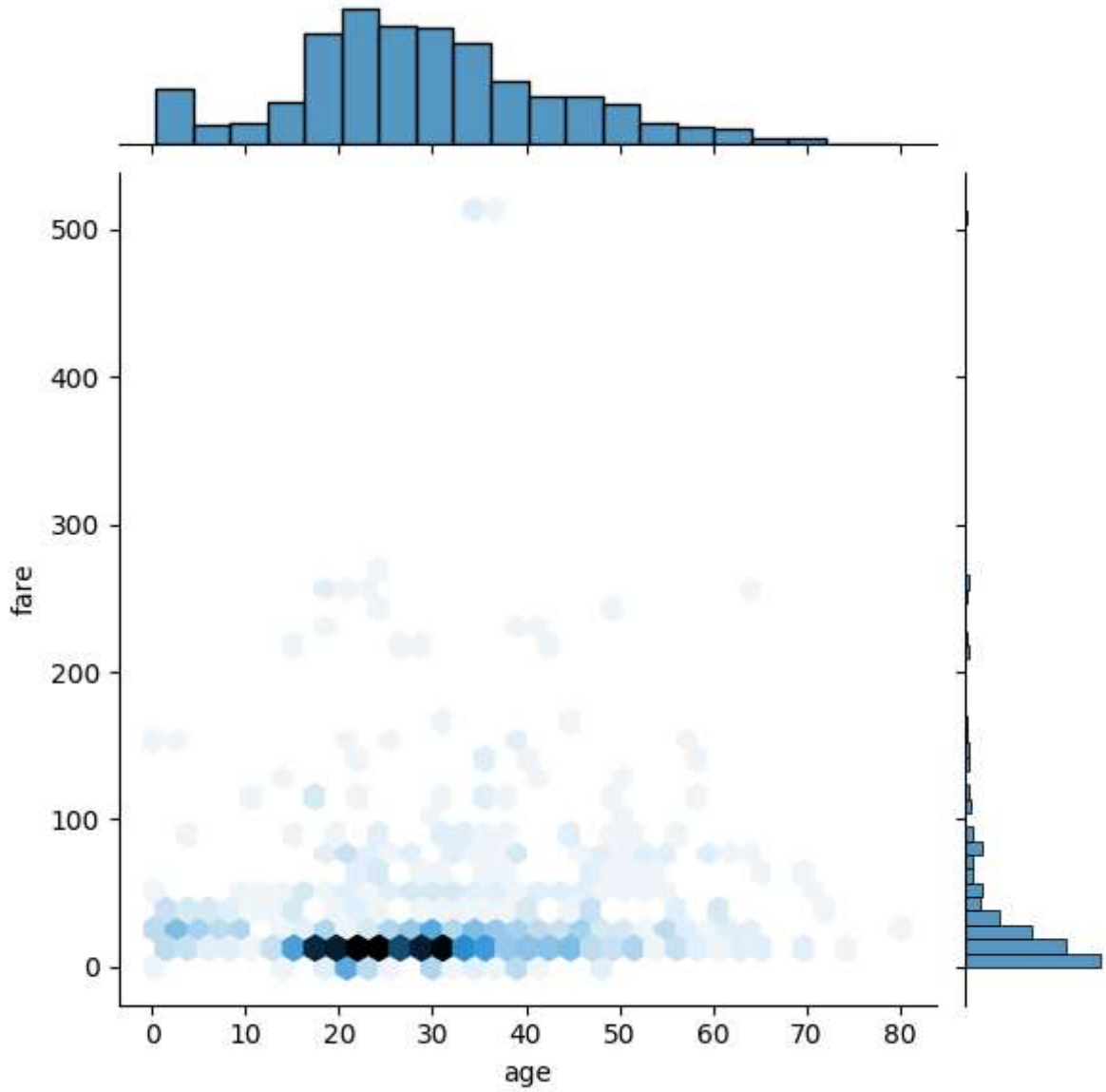
In [11]:
```
sns.histplot(df['age'], bins=10, kde=True)
plt.title("Distribution of Age")
plt.show()
```



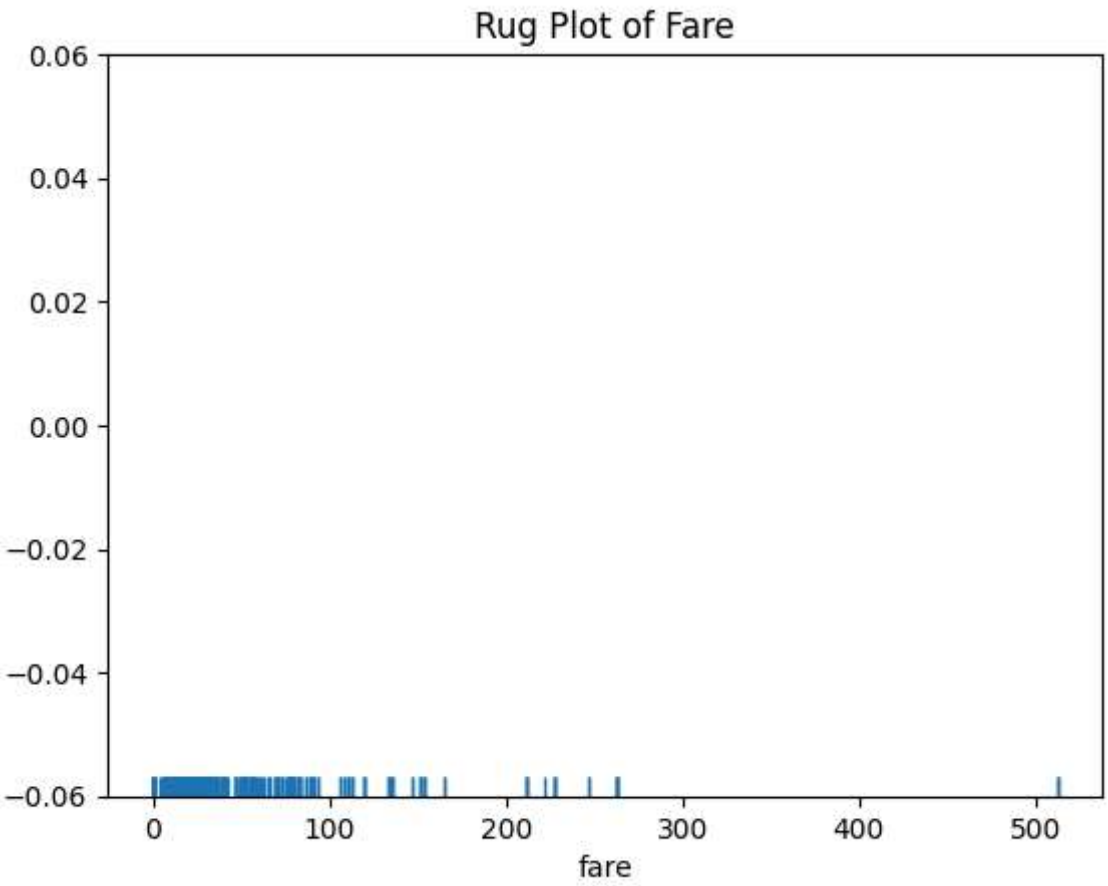
In [12]:
```
sns.jointplot(x='age', y='fare', data=df, kind='scatter')
plt.show()
```



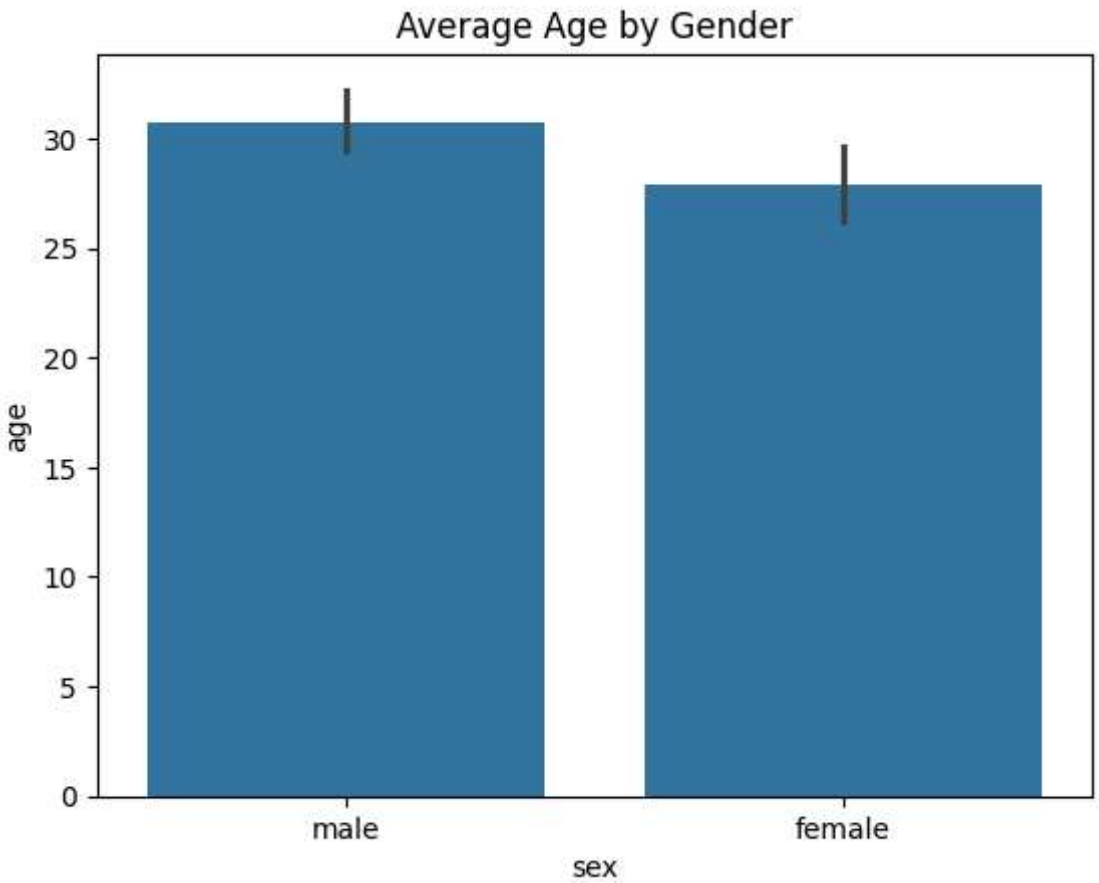
In [13]:
```
sns.jointplot(x='age', y='fare', data=df, kind='hex')
plt.show()
```



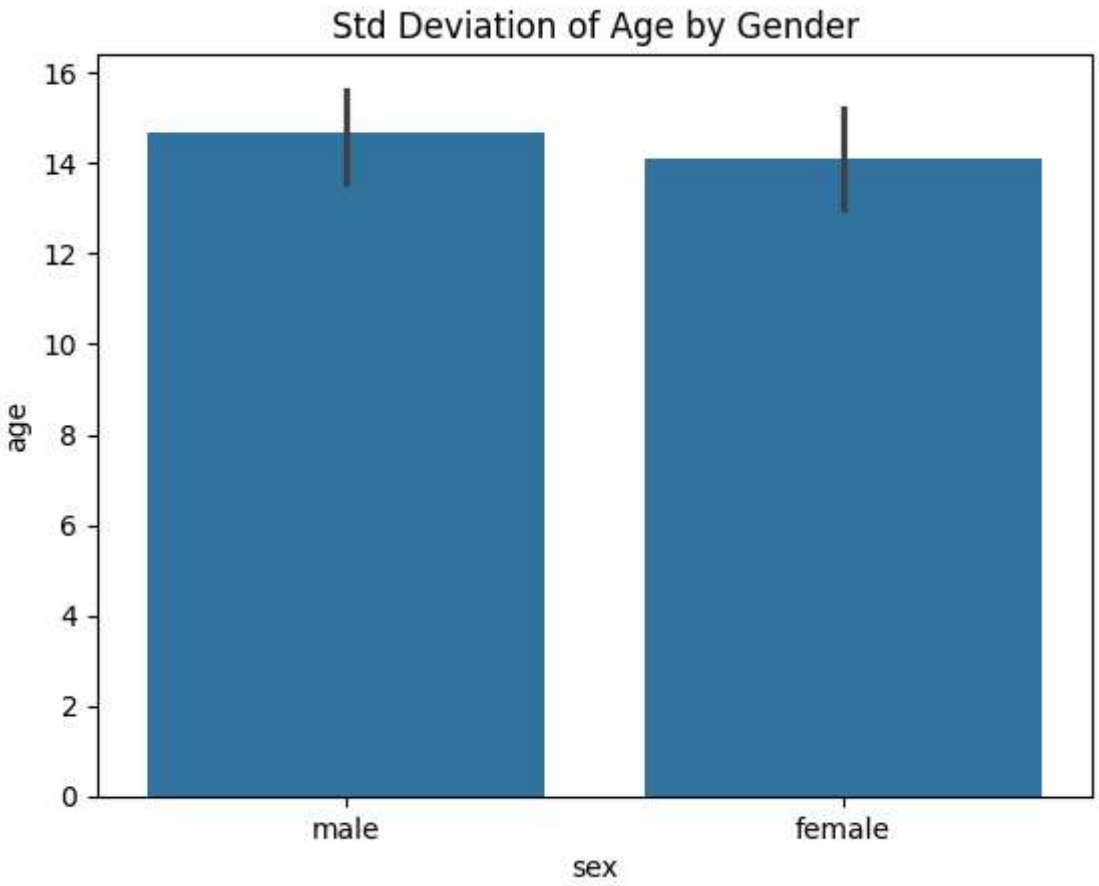
In [14]:
```
sns.rugplot(df['fare'])
plt.title("Rug Plot of Fare")
plt.show()
```



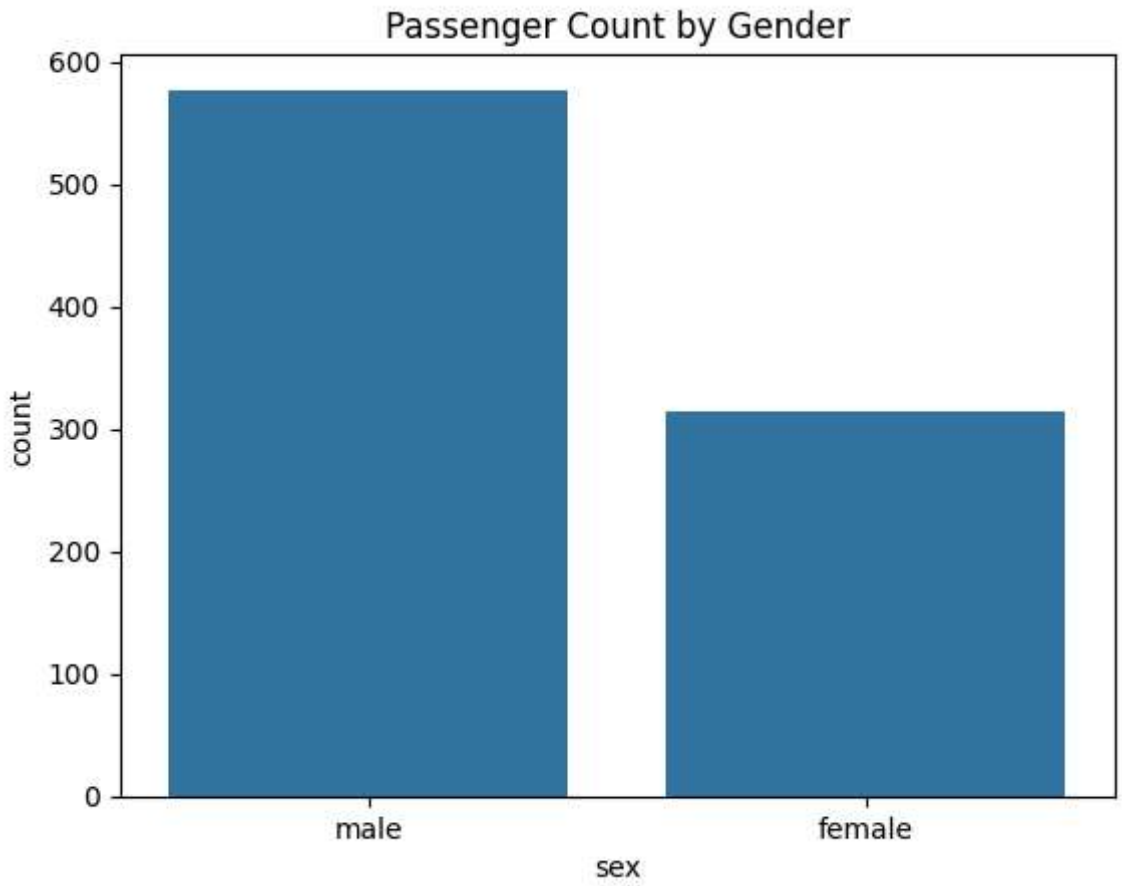
In [15]:
```
sns.barplot(x='sex', y='age', data=df)
plt.title("Average Age by Gender")
plt.show()
```



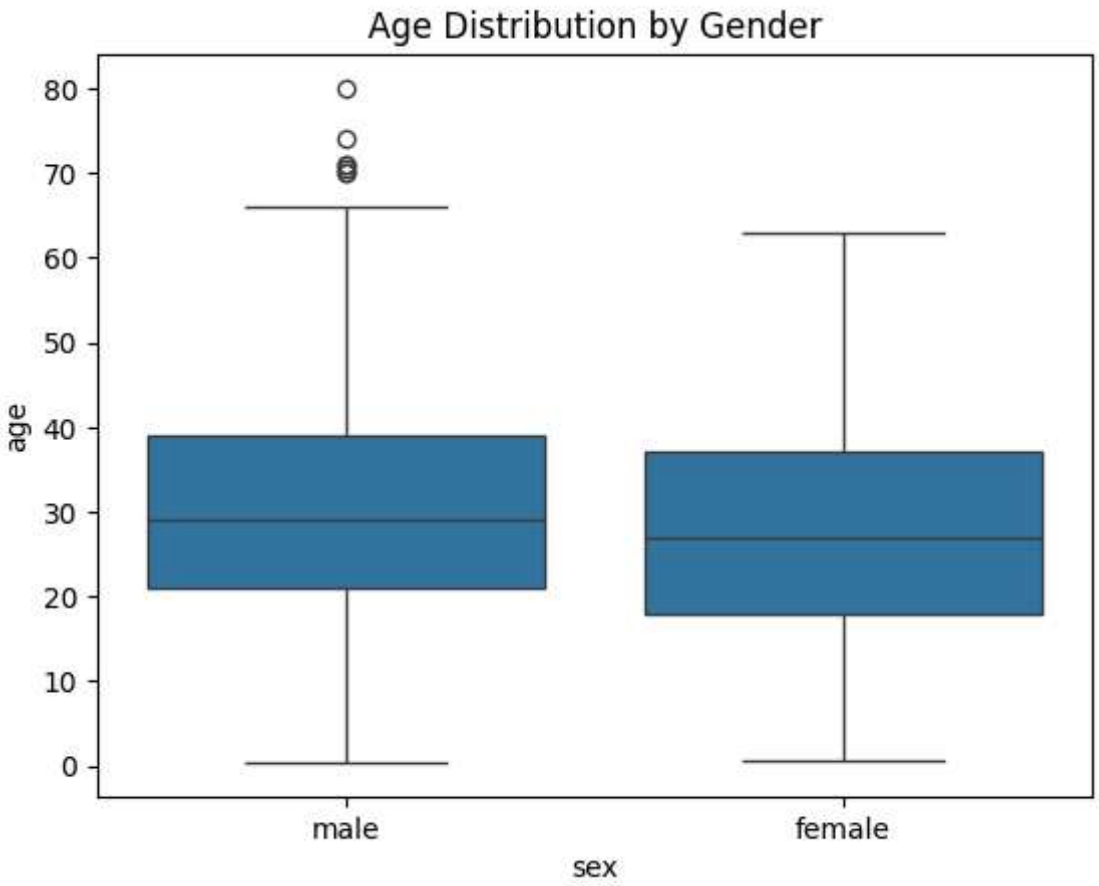
In [16]:
```
sns.barplot(x='sex', y='age', data=df, estimator=np.std)
plt.title("Std Deviation of Age by Gender")
plt.show()
```



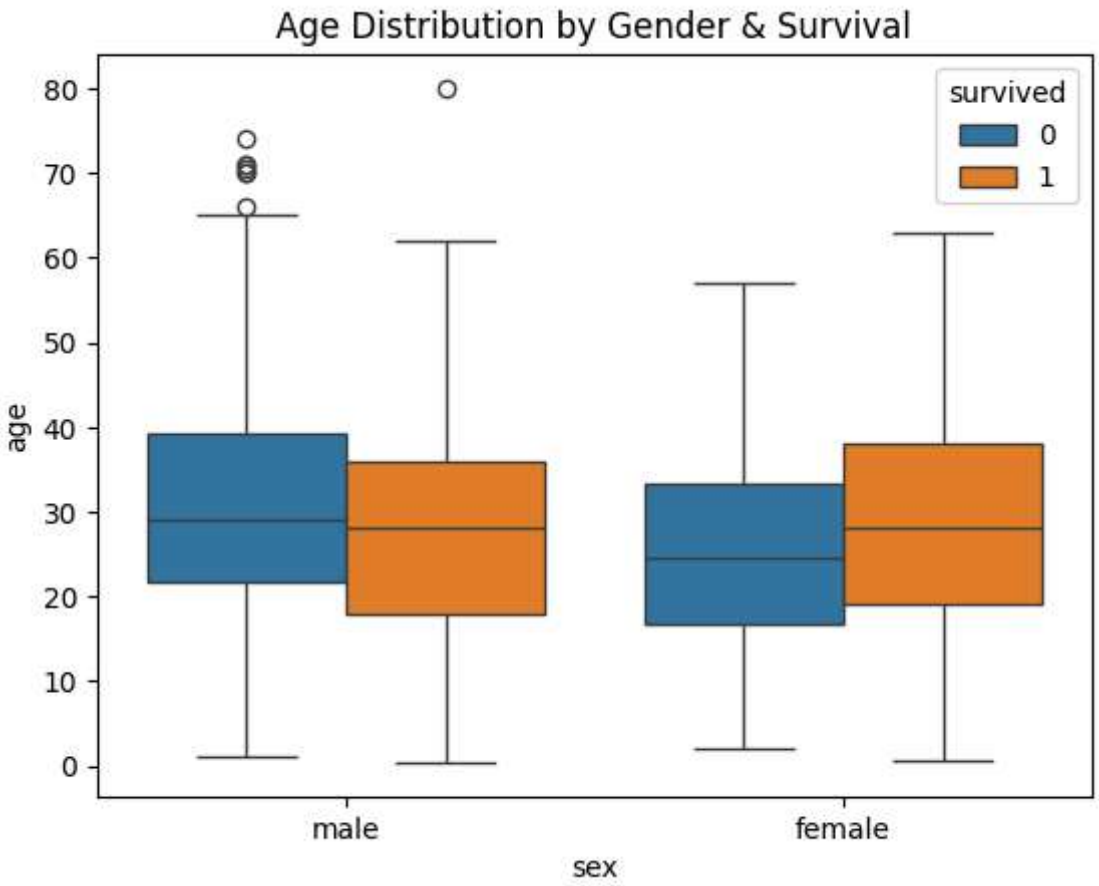
In [17]:
```
sns.countplot(x='sex', data=df)
plt.title("Passenger Count by Gender")
plt.show()
```



In [18]:
```
sns.boxplot(x='sex', y='age', data=df)
plt.title("Age Distribution by Gender")
plt.show()
```

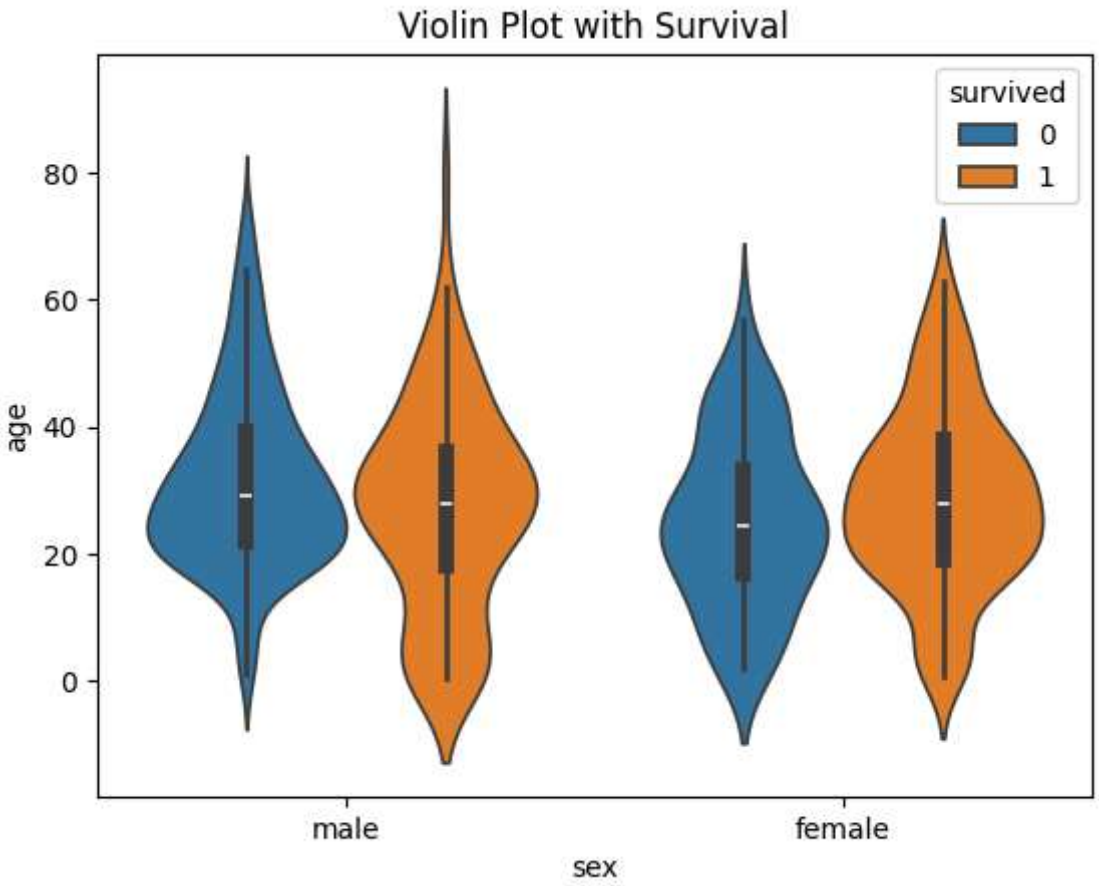


In [19]:
```
sns.boxplot(x='sex', y='age', hue='survived', data=df)
plt.title("Age Distribution by Gender & Survival")
plt.show()
```

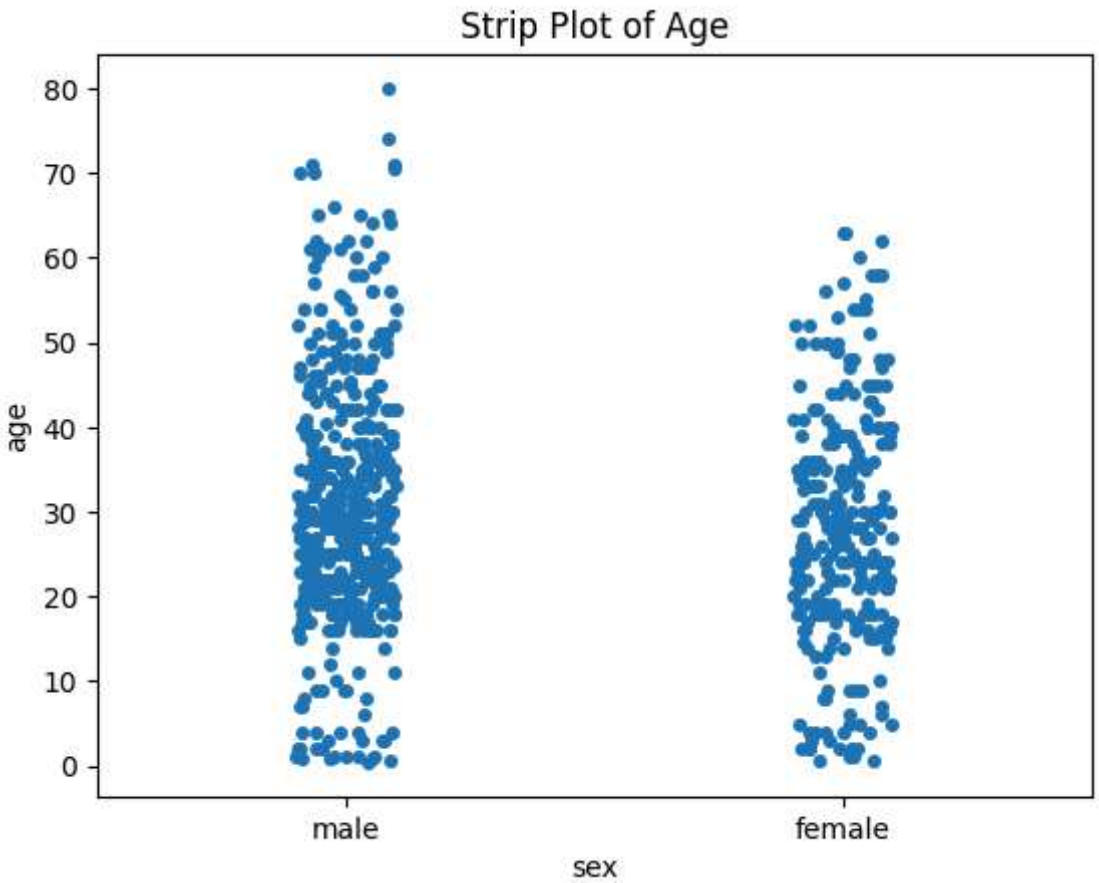


In [20]:
```
sns.violinplot(x='sex', y='age', data=df)
plt.title("Violin Plot of Age by Gender")
plt.show()
```

In [21]:
```
sns.violinplot(x='sex', y='age', hue='survived', data=df)
plt.title("Violin Plot with Survival")
plt.show()
```

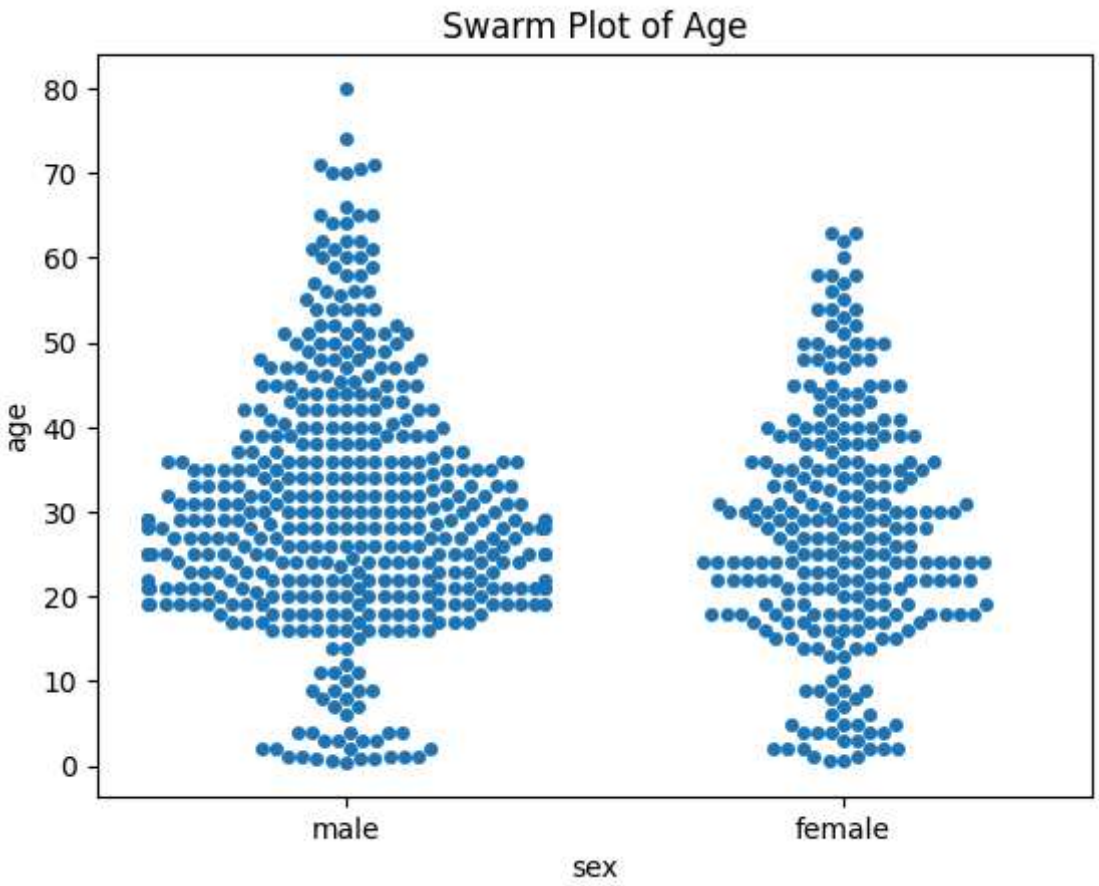


In [22]:
```
sns.stripplot(x='sex', y='age', data=df, jitter=True)
plt.title("Strip Plot of Age")
plt.show()
```

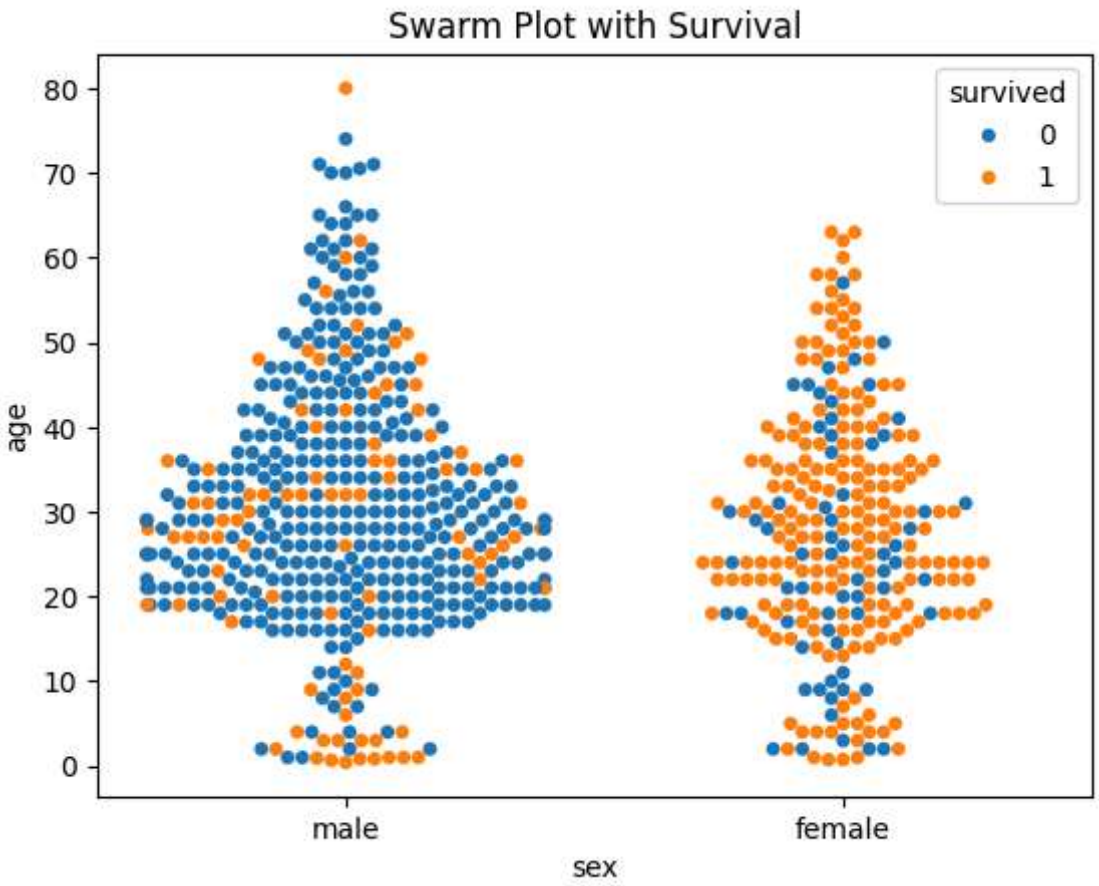


In [23]:
```
sns.stripplot(x='sex', y='age', hue='survived', data=df, jitter=True)
plt.title("Strip Plot with Survival")
plt.show()
```

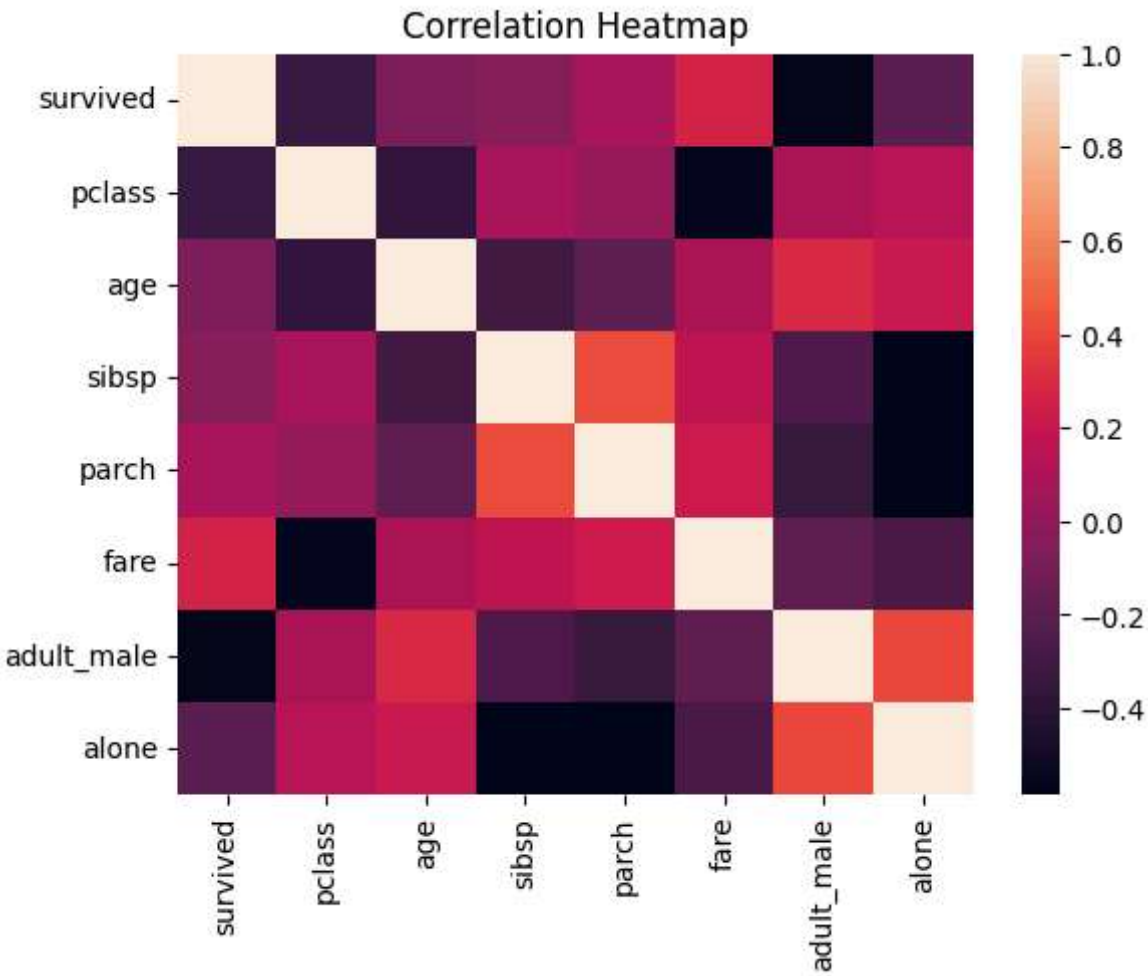
In [24]:
```
sns.swarmplot(x='sex', y='age', data=df)
plt.title("Swarm Plot of Age")
plt.show()
```



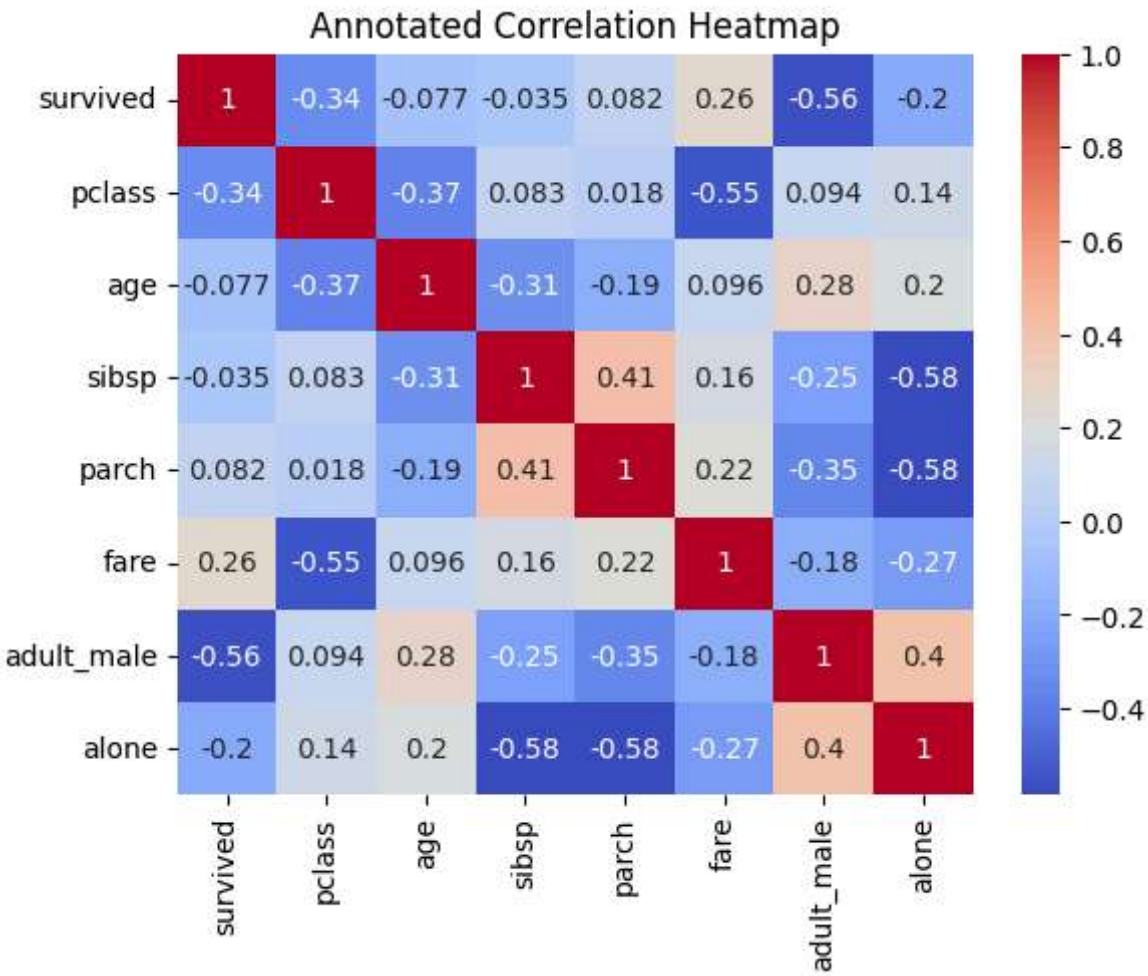
In [25]:
```
sns.swarmplot(x='sex', y='age', hue='survived', data=df)
plt.title("Swarm Plot with Survival")
plt.show()
```



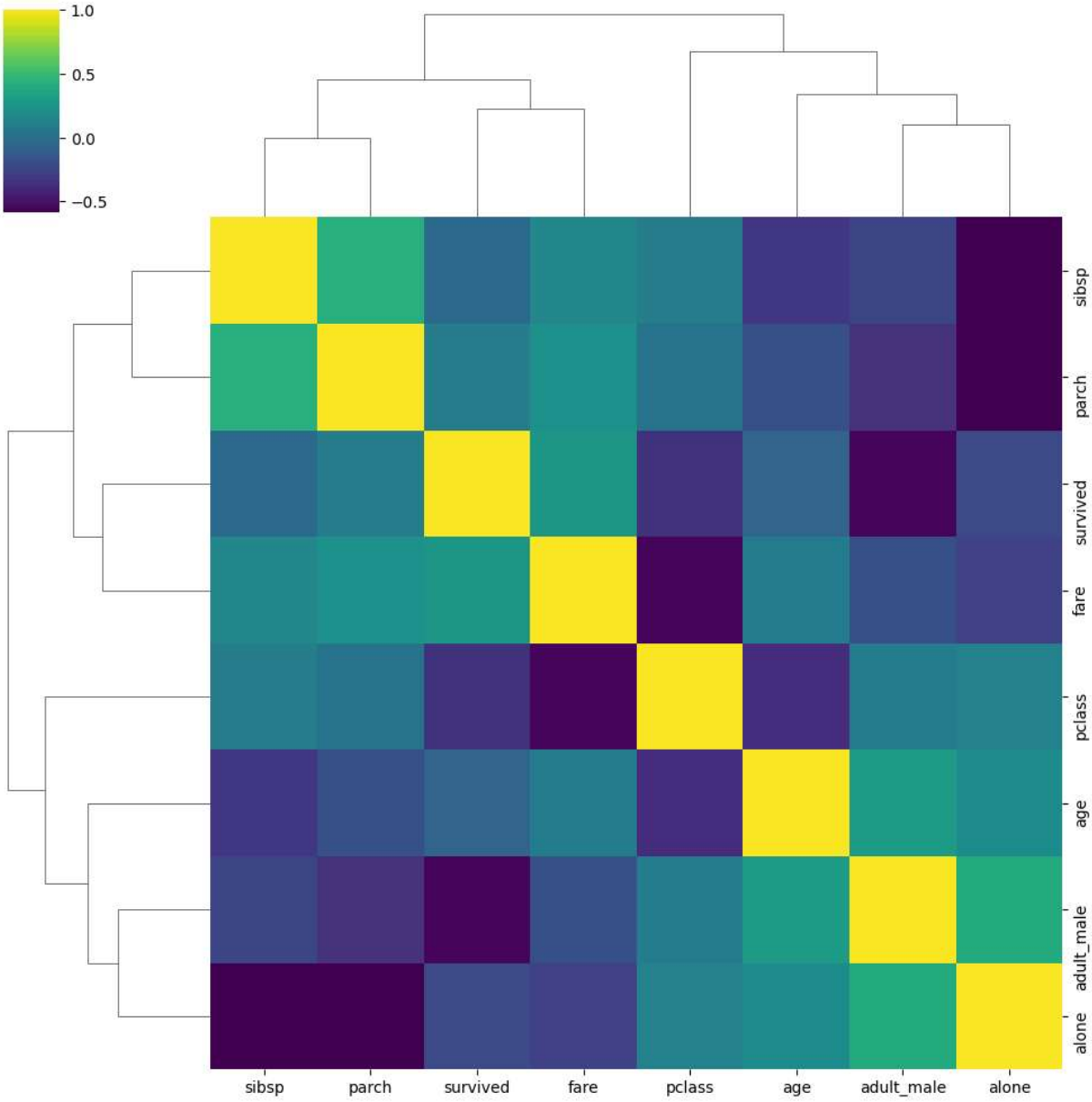
In [26]:
```
corr = df.corr(numeric_only=True)

sns.heatmap(corr)
plt.title("Correlation Heatmap")
plt.show()
```



In [27]:
```
sns.heatmap(corr, annot=True, cmap="coolwarm")
plt.title("Annotated Correlation Heatmap")
plt.show()
```



In [28]:
```
sns.clustermap(corr, cmap="viridis")
plt.show()
```


1.0
0.5
0.0
−0.5
sibsp
parch
survived
fare
pclass
age
adult_male
alone